

3. (a) Crude oil can be separated into simpler mixtures called fractions. These fractions contain hydrocarbon compounds called alkanes. **Table 1** shows information about some of the fractions obtained from crude oil by fractional distillation.

Fraction	Boiling point range (°C)	Number of carbon atoms present in the alkanes
petroleum gases	< 20	C ₁ -C ₄
petrol	30-75	C ₅ -C ₁₀
naphtha	70-170	C ₈ -C ₁₂
kerosene	170-250	C ₁₀ -C ₁₄
diesel oil	250-340	C ₁₄ -C ₂₄
lubricating oil	340-500	C ₂₁ -C ₃₀
fuel oil	490-580	C ₂₅ -C ₃₅
residue	> 580	>C ₃₅

Table 1

Use **only** the information in **Table 1** to answer parts (i)-(iii).

- (i) Hexane has a boiling point of 68 °C. Give the name of the fraction which contains hexane. [1]

.....

- (ii) One alkane is found in kerosene and in diesel. Give the number of carbon atoms in this alkane. [1]

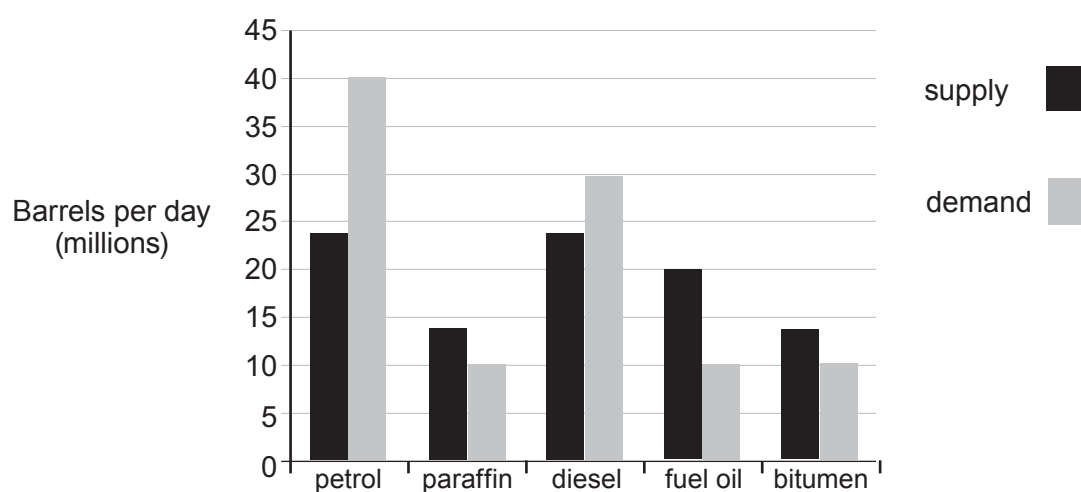
.....

- (iii) Give the number of carbon atoms in the alkane which has the **lowest** boiling point. [1]

.....



(b) The bar chart shows the **supply** and **demand** for some crude oil products.



Name the fractions for which the demand is greater than the supply. Suggest a reason why these fractions are in high demand. [2]

Fractions and

Reason

.....

